

The van der Pol oscillator, the mathematical model

The forced Van der Pol oscillator:

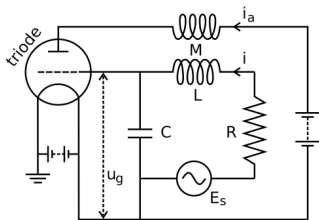
$$\frac{d^2x}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x - A \sin(\omega t) = 0,$$

has two linear and one non-linear parameters: μ, A, ω , where A is the amplitude, of the wave function and ω is its angular velocity.

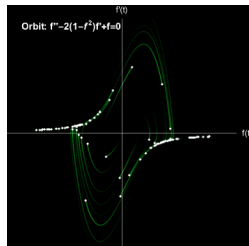
The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 \mapsto \ddot{x}, \dot{x}, x$$

and the time t .

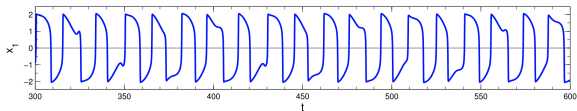


Electrical circuit resulting in a forced Van der Pol oscillator

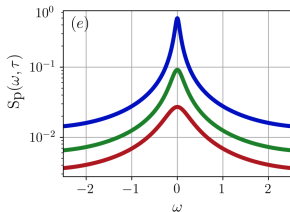
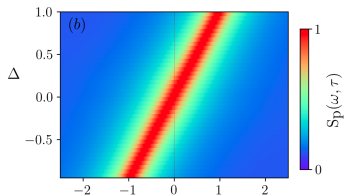
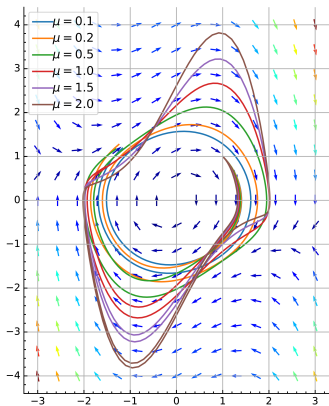


Randomly chosen initial conditions are attracted to a stable orbit

Three spaces to observe the Van der Pol oscillator



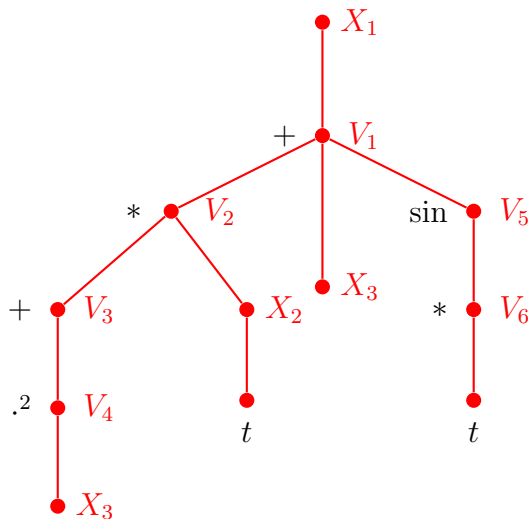
Chaotic behaviour in the Van der Pol oscillator with sinusoidal forcing.



Phase portrait of the Van der Pol's equation

Oscillator power spectra with two parameters

The causality graph of the Van der Pol oscillator model



$$\frac{d^2x}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x - A \sin(\omega t) = 0,$$

$$X_1(t) = \mu(1 - X_3^2) * X_2(t) - X_3 + A \sin(\omega * t).$$

The adjacent matrix Z

$$X_1(t) = \mu(1 - X_3^2) \times X_2(t) - X_3 + A \sin(\omega t).$$

	X_1	X_2	X_3	V_1	V_2	V_3	V_4	V_5	V_6	t
X_1				1						1
X_2										1
X_3										
V_1			1		1			1		
V_2		1				1				
V_3							1			
V_4			1							
V_5									1	
V_6										1

The basic functions for the Van der Pol model

The nodes: $X_1(t), X_2(t), X_3$ are the observable variables.

The latent variables $V_1, \dots, V_6$ are in the set (the dictionary)

$$\{\sin(\cdot), (\cdot)^2, +(w_1, \dots, w_{k+1}), \times(\dots, w_0)\}.$$

The function « $+$ » has $k + 1$ linear parameters and k arguments.

The function « $\times$ » has one parameters and k' arguments.

The causal model for the graph G

$$f_G(\mathbf{w}) : \mathbf{x}_t \mapsto \hat{\mathbf{x}}_t \quad \text{with observable variables} \quad (X_2, X_3)(t) \mapsto \hat{X}_1(t)$$

with parameters $\mathbf{w}$ and data $\mathbf{x}$ has the form

$$X_1(t) = w_1(w_4 - w_5 X_3^2) \times X_2(t) - w_2 X_3 + w_3 \sin(w_6 t).$$